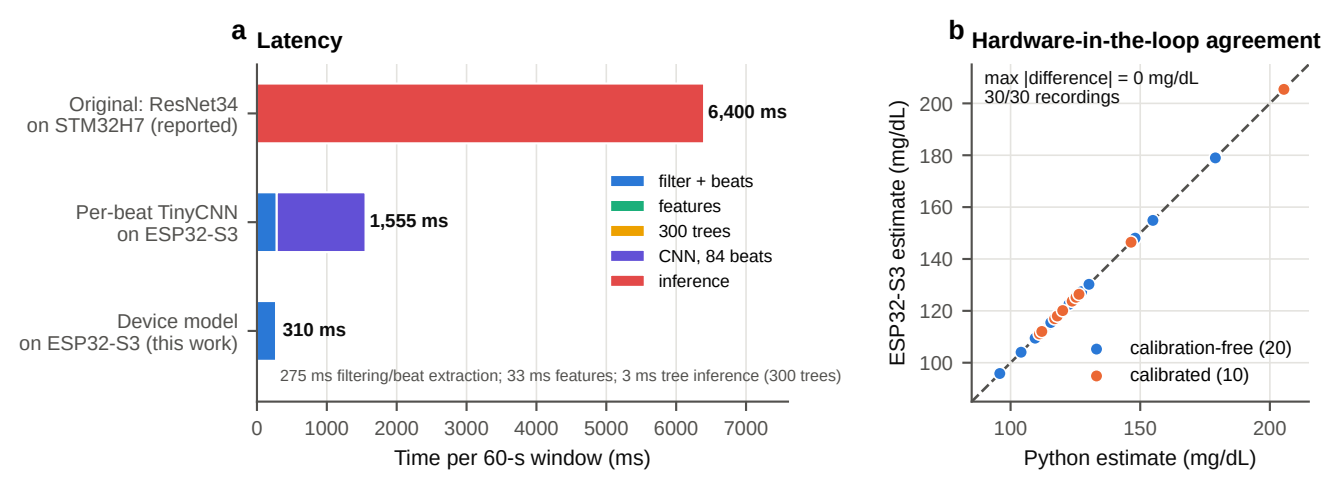
a
Latency
Original: ResNet34
on STM32H7 (reported)
6,400 ms
Per-beat TinyCNN
on ESP32-S3
1,555 ms
Device model
on ESP32-S3 (this work)
310 ms
275 ms filtering/beat extraction; 33 ms features; 3 ms tree inference (300 trees)
filter + beats
features
300 trees
CNN, 84 beats
inference
0
1000
2000
3000
4000
5000
6000
7000
Time per 60-s window (ms)
b
Hardware-in-the-loop agreement
max |difference| = 0 mg/dL
30/30 recordings
ESP32-S3 estimate (mg/dL)
100
120
140
160
180
200
100
150
200
Python estimate (mg/dL)
calibration-free (20)
calibrated (10)